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Higher MI Method and Results

1.1 INTRODUCTION

Grokking is the delayed rise of test accuracy after a model has already fitted its training examples. In this chapter we explain how a Higher-MI curve is calculated from the model's output scores, and how its timing is compared with the grok epoch.

The calculation reduces the 113 answer scores to three quantities. At every saved checkpoint, it estimates the O-information of these quantities across the test examples. The checkpoint with the largest value is taken as the predictor event.

In the five runs considered here, this event occurs after the grok epoch in every run. The early-warning fraction, the rank correlation and the mean relative lead all fail their required conditions. This conclusion applies to this benchmark adaptation.

1.2 THE TASK AND THE GROK EPOCH

The model is trained to add two integers modulo 113. If the inputs are a and b, the correct answer c is

$$c = (a + b) \bmod 113. \tag{1.1}$$

Both inputs and the answer belong to $\{0, 1, \ldots, 112\}$. For example,

$$(23 + 45) \bmod 113 = 68, \qquad (100 + 20) \bmod 113 = 120 - 113 = 7.$$

There are $113^2 = 12{,}769$ possible input pairs. The random split used for each seed is as follows.

Set	Number of pairs	Use
Training	3,830 (30%)	Used to update the model's weights.
Test	8,939 (70%)	Held aside to measure performance on unseen inputs.

Training accuracy is the fraction of training pairs answered correctly. Test accuracy is the fraction of test pairs answered correctly. High training accuracy alone does not show that the model has learned the rule.

One epoch is one pass through all 3,830 training pairs. The model is trained for 40,000 epochs. Its state is saved every 100 epochs, giving 401 checkpoints from epoch 0 through epoch 40,000. Five random seeds, numbered 0 to 4, are used. A seed fixes the initial weights and the train-test split.

The grok epoch G is the first epoch at which test accuracy exceeds 0.9:

$$G = \min\{t : \mathrm{Acc}_{\mathrm{test}}(t) > 0.9\}. \tag{1.2}$$

The observed values are

Seed	0	1	2	3	4
G	12,987	6,921	9,510	11,019	23,747

The values of G differ considerably. A useful warning should occur before G in each run and should occur later in

a run whose G is later.

1.3 LOGITS AND THREE SUMMARIES

For one input pair, the network produces a real score for each of the 113 possible answers. This score is called a *logit*. A logit is not a probability. It may be negative or greater than one, and the logits need not sum to one. The answer with the largest logit is the model's prediction. Softmax is not applied in this calculation.

Let c be the correct answer and let $\ell[j]$ denote the logit for answer j. Let r be the wrong answer with the largest logit:

$$r = \arg\max_{j \neq c} \ell[j]. \tag{1.3}$$

The three summaries are defined by

$$F_0 = \ell[c], \tag{1.4}$$

$$F_1 = \ell[r] = \max_{j \neq c} \ell[j], \tag{1.5}$$

$$F_2 = \frac{1}{111} \sum_{j \notin \{c, r\}} \ell[j]. \tag{1.6}$$

Thus, F_0 is the correct-answer score, F_1 is the largest wrong-answer score, and F_2 is the mean of the remaining 111 wrong-answer scores. The output entry for the token = is not an answer class and is excluded.

The prediction is correct precisely when the correct-answer logit is greater than every wrong-answer logit. Hence

$$\text{prediction is correct} \quad \Longleftrightarrow \quad F_0 > F_1. \tag{1.7}$$

The difference

$$m = F_0 - F_1 \tag{1.8}$$

is called the *margin*. A positive margin means a correct prediction; a negative margin means an incorrect prediction. A zero margin is a tie.

Example 1. For the test input $(23, 45)$ in seed 0, the correct answer is 68. At epoch 12,900 the reported margin is +5.4, and the runner-up is class 54. State whether the prediction is correct.

Solution. The margin is defined by $m = F_0 - F_1$. The reported value is

$$m = +5.4 > 0.$$

Therefore $F_0 > F_1$, and the model predicts the correct answer. The source report gives the margin and runner-up for this example, but not the separate values of F_0 and F_1.

For the test input $(23, 45)$, the correct answer is 68. In seed 0, its margin is about -36.4 at epoch 5,000, +5.4 at epoch 12,900, and +15.8 at epoch 20,000. At epoch 12,900, class 54 is the runner-up. These values illustrate one example only; they do not describe the full test set.

The reduction from 113 logits to three summaries is a choice made for this benchmark. It discards the identities of most wrong answers and the detailed distribution of their scores. A different reduction could give a different information curve.

1.4 MUTUAL INFORMATION AND O-INFORMATION

Mutual information $I(X;Y)$ measures the statistical dependence between X and Y. If X and Y are independent, their mutual information is zero. If knowing X helps to predict Y, it is positive. With natural logarithms, it is measured in nats. It can measure non-linear dependence as well as linear dependence.

Conditional mutual information $I(X;Y \mid Z)$ measures the dependence between X and Y when Z is already known. It tells us how much information about Y is still supplied by X after Z is taken into account.

Three variables may have overlapping information, called *redundancy*, or information that is available only when they are considered together, called *synergy*. For example, let $Z = X \oplus Y$, where $\oplus$ denotes exclusive OR on two binary values. Neither X nor Y alone determines Z, but the pair (X, Y) does.

For the three summaries, O-information is defined as

$$\Omega(F_0, F_1, F_2) = I(F_0; F_1) - I(F_0; F_1 \mid F_2). \tag{1.9}$$

Under the sign convention used here, positive values indicate that redundancy dominates, while negative values indicate that synergy dominates. The sign does not indicate whether the model has grokked. O-information describes the statistical relation among the three scores across examples; it is not a measure of accuracy.

At epoch 20,000, for example, F_0 and F_1 have correlation 0.998 across examples, while O-information is about 0.36 nats. Correlation and O-information measure different properties.

1.5 ESTIMATION BY THE KSG METHOD

The probability distributions of the three summaries are not known. We have a finite sample of 8,939 triplets at each checkpoint. A histogram estimate depends on the chosen bin widths. The report instead uses the nearest-neighbour method of Kraskov, Stögbauer and Grassberger, usually called the KSG estimator.

The number of neighbours is $k = 4$. For each sample point, the estimator finds the distance to its fourth-nearest other point in the joint space of (F_0, F_1, F_2). It uses the maximum norm: the distance between two points is the largest absolute difference among their coordinates. It then counts points strictly inside the same radius in the smaller spaces.

Step 1. Find the distance from each triplet to its fourth-nearest neighbour in the three-dimensional space.

Step 2. Use this distance as the radius around that triplet.

Step 3. Count the other points strictly inside that radius in each required coordinate or pair of coordinates.

Step 4. Substitute the counts in the mutual-information and conditional-mutual-information estimates, and combine them by the O-information formula.

The count must be strict. The point that sets the fourth-neighbour distance lies on the boundary and is not counted as lying inside it. The corrected implementation uses strict counts and was checked against a direct calculation.

Let T be the number of sample points, and let ψ denote the digamma function. Angle brackets denote the average over all sample points. The estimator forms used in the calculation are

$$I(X;Y) = \psi(k) + \psi(T) - \left\langle \psi(N_x + 1) + \psi(N_y + 1) \right\rangle, \tag{1.10}$$

$$I(X;Y \mid Z) = \psi(k) + \left\langle \psi(N_z + 1) - \psi(N_{xz} + 1) - \psi(N_{yz} + 1) \right\rangle. \tag{1.11}$$

Here N_x and N_y count points in the X and Y coordinates, respectively. The counts N_{xz} and N_{yz} are made in the corresponding two-dimensional spaces.

The exact O-information expression is symmetric in its three variables. The finite-sample estimate may depend

slightly on their order. Therefore, the estimator is applied to all six orderings of (F_0, F_1, F_2), and the six values are averaged. At seed 0 and epoch 12,900, this mean is 0.4044 nats; the standard deviation across the six orderings is 0.0099 nats.

1.6 POPULATION O-INFORMATION AND THE EVENT

At a checkpoint t, each test example gives one triplet (F_0, F_1, F_2). The 8,939 triplets give one population value, written $\Omega_{\text{pop}}(t)$. Repeating the calculation at all 401 checkpoints gives one curve for each seed. The fraction of examples satisfying $F_0 > F_1$ is exactly the test accuracy.

The principal calculation samples across examples at one fixed checkpoint. It does not measure how the summaries change over training time. The separate exploratory temporal control first averages the summaries over the test set and then studies those means in windows of training time.

The event used for scoring is the checkpoint with the global maximum:

$$P = \arg\max_t \Omega_{\text{pop}}(t). \tag{1.12}$$

This event definition was fixed before scoring. The question is whether P occurs before G, and whether the values of P follow the variation in G across seeds.

The observed curves have repeated rises and falls, both before and after grokking. In every seed, Ω_{pop} first exceeds 0.2 at epoch 1,900. The global maximum occurs after G in every run. The peak size also varies: seed 4 rises from about 0.04 to 1.80, while seed 1 changes from about 0.033 to 0.070. The common rise at epoch 1,900 may be related to general training dynamics, but its cause was not tested. The initial values also do not reproduce the strongly negative values described in the source paper.

1.7 PEAK TIMES AND THE SCORING RULES

The following table gives the grok epoch G, the peak epoch P, and the derived timing values for each seed.

Seed	G	P	$P - G$	P/G	$(G - P)/G$
0	12,987	15,500	2,513	1.19	-0.1935
1	6,921	8,500	1,579	1.23	-0.2281
2	9,510	20,600	11,090	2.17	-1.1661
3	11,019	12,900	1,881	1.17	-0.1707
4	23,747	32,500	8,753	1.37	-0.3686

For seed 0,

$$P - G = 15{,}500 - 12{,}987 = 2{,}513.$$

Thus, the peak is 2,513 epochs late. Its relative lead is

$$\frac{G - P}{G} = \frac{12{,}987 - 15{,}500}{12{,}987} \approx -0.1935.$$

A negative relative lead means that the event occurs after grokking.

The benchmark uses three fixed conditions.

(i) The early-warning fraction is the fraction of seeds for which $P < G - 100$. It must be at least 0.80, so at least four of the five seeds must lead.

(ii) Spearman's rank correlation between the five G values and the five P values must be at least 0.90.

(iii) The mean of $(G - P)/G$ across the five seeds must be from 0.05 to 0.90.

The ranks of G, in seed order, are $(4, 1, 2, 3, 5)$. The ranks of P are $(3, 1, 4, 2, 5)$. The rank differences are $(1, 0, -2, 1, 0)$, and the sum of their squares is 6. For $n = 5$, Spearman's rank correlation is

$$\rho = 1 - \frac{6 \sum d_s^2}{n(n^2 - 1)} = 1 - \frac{6(6)}{5(25 - 1)} = 1 - \frac{36}{120} = 0.70. \tag{1.13}$$

The mean relative lead is -0.4254. The results are summarised below.

Condition	Required	Observed	Result
Early-warning fraction	≥ 0.80	$0/5 = 0$	Fail
Spearman correlation	≥ 0.90	0.70	Fail
Mean relative lead	0.05 to 0.90	-0.4254	Fail

Conclusion. All three conditions fail. None of the five global maxima occurs before grokking.

1.8 TEMPORAL CONTROL AND LIMITATIONS

In the exploratory temporal control, the three summaries are first averaged over the test set at each checkpoint. The mean triplets are then collected in windows of 100 consecutive checkpoints. Each window spans about 9,900 epochs from its first to last checkpoint, and the window advances by 10 checkpoints. This gives 31 values centred from epoch 5,000 to epoch 35,000. No temporal value is available before epoch 5,000.

In seed 0, the temporal curve begins near 1.55, falls to about 0.52 near epoch 10,000, and later remains near 1.2 to 1.45. The temporal curve is not negative in any seed. The population curve does not show the same dip. These calculations use different samples and answer different questions. The wide windows do not provide a precise warning time.

A possible concern in the population calculation is shared example difficulty. An easy example may have high F_0 and low F_2, while a difficult example may have F_0 close to F_1. This can make the summaries move together even if grokking is not the cause. The temporal comparison does not remove this concern.

Only five seeds are available. With five values, Spearman's correlation can take only 11 distinct values. Checkpoints are 100 epochs apart, so a faster change may be missed. The three-score reduction discards information from the 113 logits. The common rise at epoch 1,900 remains unexplained.

Further work may use more seeds, save checkpoints more densely near epoch 1,900 and near each grok epoch, test other reductions of the logits, and control for example difficulty.

1.9 COMPARISON WITH THE SOURCE PAPER

The source paper by Pomarico et al. (2025) studied a matrix product state classifier on three-class tasks. It reported an O-information peak near grokking in that setting. The present benchmark uses a one-layer transformer on modular addition with 113 answer classes. It uses the project-defined summaries (F_0, F_1, F_2) and estimates information across test examples at fixed checkpoints.

The source paper did not claim that the peak gives an early warning. The model, task, variables and sampling procedure differ. Therefore, the result here does not refute the source paper. It shows that the chosen global-maximum rule does not give an early warning in this benchmark.

1.10 REVIEW QUESTIONS

1. What does each of F_0, F_1, F_2 represent?
2. Under what condition is the model's prediction correct?

3. What does a positive or negative O-information value mean under the sign convention used here?
4. How does the population calculation differ from the temporal control?
5. Why does the global-maximum event fail the early-warning test in these five runs?